

FIN ST552–ST641

Refinement-Speed No-Go, Conditional Light-Cone Emergence,
and the Two-Clock Continuum Architecture

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Six consecutive research rounds
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Scope. This report studies whether FIN can generate a layer-invariant propagation speed or causal cone from its strict twelve-state operator and exact refinement results. It includes ST552–ST641 only and preserves the legacy/strict kernel split.

Evidence boundary. All results are analytic, exact finite-dimensional, interval, or explicitly numerical within declared models. No physical light field, laboratory clock, ruler, detector record or independent external audit is supplied.

Repository. <https://github.com/hyconiek/Fractal-Nadsoliton-Theory> **License.** CC BY 4.0.

Abstract

Six consecutive FIN research rounds investigate the hypothesis that internal observers on different fractal/refinement layers may reconstruct the same propagation speed even when their local length and time standards differ. The correct layer law is

$$c_n^{\text{phys}} = \frac{\ell_n}{\tau_n} \hat{c}_n, \quad \frac{c_{n+1}^{\text{phys}}}{c_n^{\text{phys}}} = \frac{\alpha_n}{\beta_n} \frac{\hat{c}_{n+1}}{\hat{c}_n}.$$

Equal length/time scaling is sufficient only when the dimensionless dispersion slope is itself refinement invariant.

For the fixed strict C_{12} operator, a formal continuous-symbol slope $\hat{c} = 1.9015886526$ can be computed, but the first available discrete mode has phase and group speeds 1.65852 and 1.22656, and the full radial graph has no exact causal cone. Candidate refinements separate into inequivalent universality classes. A paired finite-range continuation has quadratic low-wave-number symbol and speed 1.9532829072. A positive $d^{-1.8}$ tail has fractional symbol order 0.8, wave exponent 0.4 and divergent low- k group speed. The literal signed oscillatory strict continuation loses positive-semidefinite Laplacian structure already at the tested carrier $N = 48$.

The central no-go theorem comes from the exact dyadic refinement fiber. For every $q \geq 0$, $A_{24}(q)J = JA_{12}$ and all coarse unitary and heat records are identical, while the fine formal speed obeys

$$\hat{c}(q)^2 = \hat{c}(0)^2 + 72q.$$

Thus complete coarse dynamics is exactly compatible with a continuous unbounded family of fine speeds. Across a tower,

$$C_{2N} = C_N + \frac{N^2}{2} q_N.$$

A finite infinite-depth speed requires $\sum N^2 q_N < \infty$; a uniform exponential propagation bound requires exponential rather than polynomial decay of antipodal rates. Exact speed or trace-density stationarity conditionally selects $q_N = 0$, but neither stationarity law is strict-sourced.

For the conditional zero-rate finite-range tower on a fixed physical circle, A_N/a_N^2 converges to $-C\partial_x^2$. A scalar $1 + 1$ wave equation with an exact limiting domain of dependence therefore emerges conditionally, and numerical finite-lattice tails outside $1.05\hat{c}t$ fall to 8.2×10^{-9} by $N = 12288$. However, one universal refinement clock is impossible: unitary/heat limits require $t \sim a^{-2}$ while the second-order wave channel requires $t \sim a^{-1}$. Two channel-specific clock classes and a dimensional length/time anchor remain necessary. FIN therefore permits a coherent emergent-cone completion but does not make it inevitable, does not derive the SI value of c , and does not yet supply Lorentzian $3 + 1$ spacetime or physical light.

Claim vocabulary. **[PROVEN]** means an analytic/exact/interval theorem in scope; **[STRONG EVIDENCE]** means reproducible nonexhaustive numerical evidence; **[CONDITIONAL]** requires displayed additional premises; **[OPEN]** is unresolved; **[BLOCKED]** denotes a missing typed input, not a refutation of all future completions.

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1 Executive synthesis

Main result. The current strict endpoint and exact coarse-refinement axioms do not determine a propagation speed. They admit infinitely many fine generators with identical complete coarse spectral dynamics and arbitrarily different fine speed/locality data. A conditional local continuum with a wave cone exists only after choosing a zero-rate finite-range tower, coordinate scaling, a ballistic wave clock and dimensional calibration.

Table 1: Central status map.

Question	Answer	Status
Does A_{12} define a physical c ?	It defines finite dimensionless proxies, not a continuum slope or SI speed.	[BLOCKED]
Does the full strict graph have an exact cone?	No; opposite vertices are directly coupled.	[PROVEN]
Which tail permits finite low- k speed?	Exactly a quadratic symbol $\lambda \sim Ck^2$.	[PROVEN]
Does $d^{-1.8}$ give that symbol?	A positive envelope has order 0.8, hence divergent group speed.	[PROVEN]
Does exact refinement select a speed?	No; the q fiber preserves all coarse records but changes fine speed.	[PROVEN]
Can a conditional local limit exist?	Yes, for the added zero-rate finite-range tower with A_N/a_N^2 .	[CONDITIONAL]
Can one clock preserve all channels?	No; U/P need a^{-2} and the wave channel needs a^{-1} .	[PROVEN]
Is physical light derived?	No; $3 + 1$ geometry, boosts, gauge structure and calibration remain missing.	missing

The strongest scientific conclusion is a distinction between *compatibility* and *inevitability*. FIN is compatible with a local wave completion. Current FIN does not select it over fractional, signed or nonlocal refinements.

2 Frozen strict core and channel definitions

On C_{12} define

$$W_{xx} = 0, \quad W_{xy} = K_{\text{strict}}(d(x, y)), \quad A = sI - W,$$

with

$$K_{\text{strict}}(d) = \frac{\cos(0.18575d + 0.16250)}{1 + d^{1.8}}.$$

The strict weights at distances 1 through 6 are positive and $A \succeq 0$. Three spectral calculi are kept distinct:

$$U_t = e^{-itA}, \quad P_t = e^{-tA}, \quad C_t = \cos(t\sqrt{A}).$$

U_t and P_t use the eigenvalue λ as phase frequency/decay rate; the second-order wave channel uses $\sqrt{\lambda}$. Common projectors do not imply common dimensional time maps.

For a continuous interpolation of the finite symbol,

$$\lambda_{12}(q) = 2 \sum_{d=1}^5 w_d(1 - \cos dq) + w_6(1 - \cos 6q),$$

the formal coefficient is

$$\hat{c}_{12}^2 = \sum_{d=1}^5 w_d d^2 + 18w_6, \quad \hat{c}_{12} = 1.9015886526.$$

Yet $q_{\min} = \pi/6$, so C_{12} contains no actual $q \rightarrow 0$ sequence.

3 Round I: ST552–ST566

3.1 Speed proxy and causal obstruction

The first strict mode gives

$$v_{\text{phase}} = 1.6585246837, \quad v_{\text{group}} = 1.2265636961,$$

well separated from the formal tangent. The full graph directly couples every cyclic distance, so the unitary Born probability at the opposite site begins as $O(t^2)$ and heat probability as $O(t)$. No strict finite cone is present.

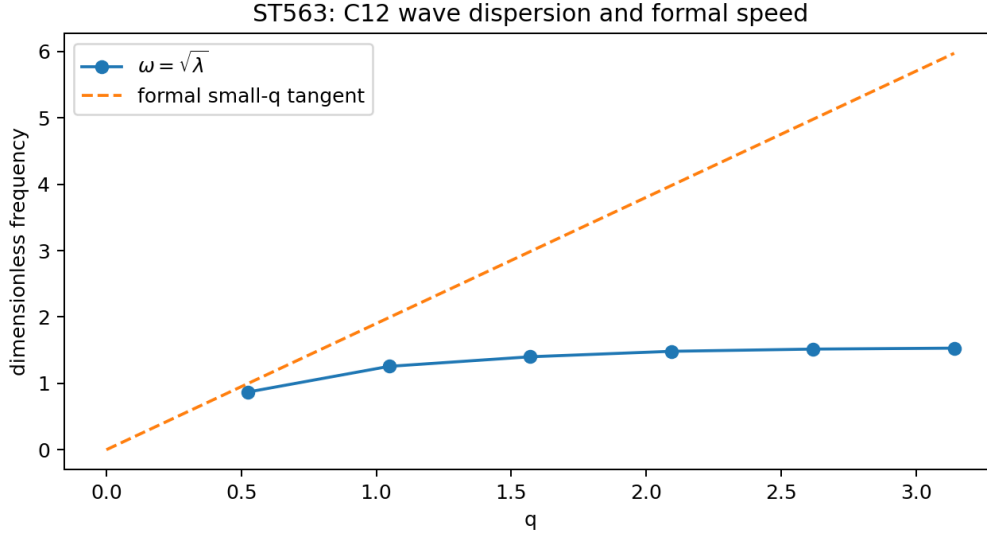


Figure 1: Finite C_{12} dispersion and its formal tangent. The tangent is not an available physical mode.

Layer covariance requires

$$\frac{\alpha_n \hat{c}_{n+1}}{\beta_n \hat{c}_n} = 1.$$

The frequently proposed condition $\alpha_n = \beta_n$ is sufficient only if $\hat{c}_{n+1} = \hat{c}_n$, which is precisely the missing refinement theorem.

3.2 Other Round-I frontiers

The 38-relation degree-eight block again failed rational reconstruction after a conditioned five-prime lift, so no characteristic-zero rank was promoted. The first-transition orbit remained globally open. A CPTP operational-conjugacy fixture passed with trace-preservation residuals below 1.5×10^{-14} . IR interval continuation reached $b = 0.2592$ before radius inflation became method-gated.

3.3 Round-I ledger

Program	Status	Result
ST552–556	open/gated	Conditioned 38-relation lift still fails rational reconstruction.
ST557–559	[STRONG EVIDENCE]	Two-level/reflection searches support the candidate transition; globality open.

Program	Status	Result
ST560	[PROVEN]	Equivariant irreducible Markov selector is uniform.
ST561	[STRONG EVIDENCE]	Synthetic three-channel finite-shot classifier passes 1% target.
ST562	[PROVEN]	CPTP operational conjugacy fixture.
ST563	[PROVEN]	Finite speed proxies and exact-cone obstruction.
ST564	[PROVEN]	IR continuation reaches $b = 0.2592$.
ST565–566	blocked	No physical cone, speed source or evidence.

4 Round II: ST567–ST581

4.1 Three inequivalent refinements

For a paired finite-range continuation,

$$\lambda(q) = 2 \sum_{d=1}^6 w_d (1 - \cos dq) = Cq^2 + O(q^4), \quad C = \sum_{d=1}^6 w_d d^2,$$

and

$$\hat{c}_{\text{FR}} = \sqrt{C} = 1.9532829072.$$

The first-mode slopes converge to this number as $N \rightarrow \infty$.

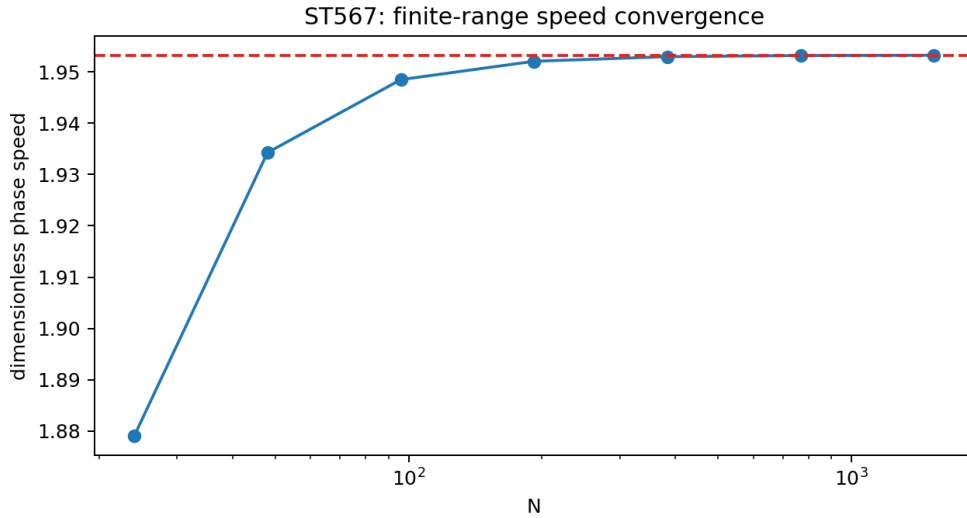


Figure 2: Conditional finite-range phase-speed convergence.

For the positive envelope $a_d = (1 + d^{1.8})^{-1}$, the fitted low- q symbol exponent is 0.7970, consistent with $\eta - 1 = 0.8$. Hence $\omega \sim |q|^{0.4}$ and $v_g \sim |q|^{-0.6}$.

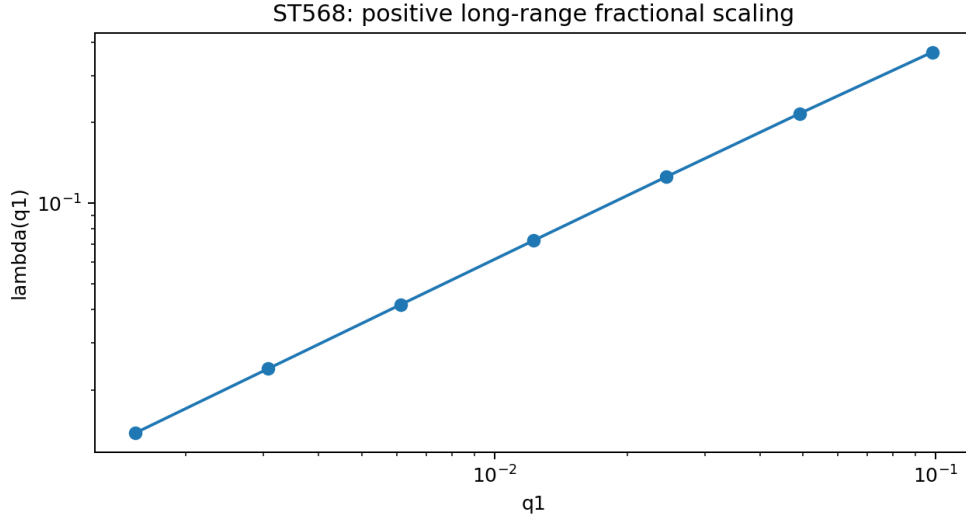


Figure 3: Positive long-range envelope: fractional rather than quadratic scaling.

The literal oscillatory strict formula, extended to all distances, introduces negative weights and negative Laplacian modes at tested $N = 48$.

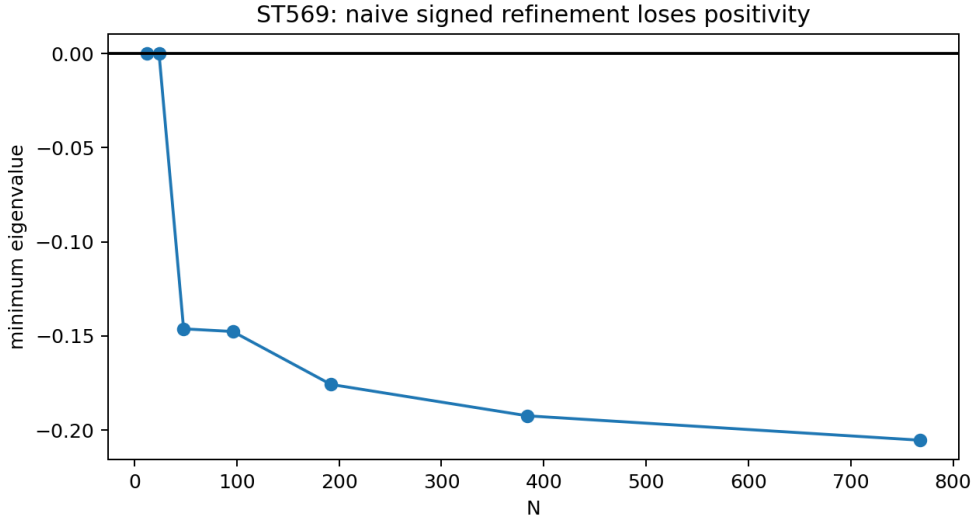


Figure 4: Naive signed strict refinement loses positive-semidefinite generator structure.

Thus A_{12} admits mutually incompatible continuations: local/quadratic, positive/fractional and signed/indefinite.

4.2 Locality and chart results

The finite-range exponential-moment velocity proxy is 3.00711, while the maximal sampled wave group velocity is 1.95328. This is an approximate single-particle cone, not Lorentz invariance. A logarithmic IR chart reduced numerical Jacobian conditioning by a median factor 35.4, but the interval implementation was not yet certified.

4.3 Round-II ledger

Program	Status	Result
ST567	[PROVEN]	Finite-range quadratic dispersion limit.
ST568	[STRONG EVIDENCE]	Positive long-range fractional exponent ≈ 0.8 .
ST569	[PROVEN]	Finite counterexample to naive positive signed refinement.
ST570	[PROVEN]	Layer-speed covariance depends on refinement.
ST571	conditional	Finite-range LR velocity proxy.
ST572–573	[PROVEN]	Spectrum/speed coordinate distinction and short-time tails.
ST574–577	gated	Transition interval and characteristic-zero methods remain open.
ST578	[CONDITIONAL]	Three-channel noise scaling.
ST579	[STRONG EVIDENCE]	Log-chart floating conditioning improved.
ST580–581	blocked	No physical c , Lorentz cone or evidence.

5 Round III: ST582–ST596

5.1 Dispersion theorem

Theorem 5.1 (Positive-tail symbol). *If $a_d \sim cd^{-1-\alpha}$ with $0 < \alpha < 2$, then*

$$2 \sum_{d \geq 1} a_d (1 - \cos kd) \sim 2cI_\alpha |k|^\alpha, \quad I_\alpha = \frac{\pi}{2\Gamma(1+\alpha) \sin(\pi\alpha/2)}.$$

For $\alpha = 0.8$, $2I_\alpha = 3.5466218138$.

Theorem 5.2 (Speed trichotomy). *If $\lambda(k) \sim C|k|^\nu$, then*

$$v_g(k) \sim \sqrt{C} \frac{\nu}{2} |k|^{\nu/2-1}.$$

A finite nonzero low- k speed exists if and only if $\nu = 2$.

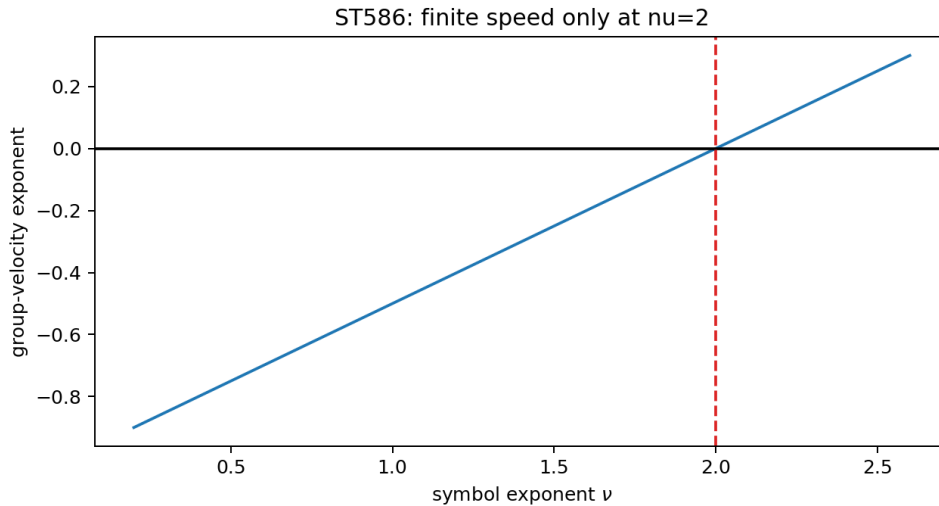


Figure 5: Only a quadratic generator symbol yields a finite nonzero massless speed.

For finite range, exponential conjugation gives the single-particle bound

$$|\langle x | e^{-itA} | y \rangle| \leq e^{-\mu(r-v_\mu|t|)}, \quad v_\mu = \frac{2}{\mu} \sum_d w_d \sinh(\mu d).$$

Long-range direct couplings instead give immediate polynomial spatial tails.

5.2 Honest IR failure

Direct interval substitution $x = e^z$ inflated dependencies. The first proposed log Krawczyk cell has margin -0.00663 and is rejected. No log interval chain or degree theorem is claimed. This is a bounded no-go for the naive chart, not disappearance of the numerical root.

5.3 Round-III ledger

Program	Status	Result
ST582–584	bounded stop	Naive log interval chart rejected; no degree promotion.
ST585	[PROVEN]	Positive-tail fractional-symbol theorem.
ST586	[PROVEN]	Dispersion-exponent speed trichotomy.
ST587–589	[PROVEN]	Finite-range cone bound, tail classification and scale torsor.
ST590	[CONDITIONAL]	Dimensionless speed estimator.
ST591–593	gated	Transition/reflection/algebra methods frozen.
ST594–596	blocked	No seed/gain/selector, physical c or evidence.

6 Round IV: ST597–ST611

6.1 Exact speed fiber and observer blindness

The known exact dyadic family satisfies

$$A_{24}(q)J = JA_{12}, \quad q \geq 0.$$

Its even modes and every coarse spectral function are independent of q , while

$$\boxed{\widehat{c}(q)^2 = \widehat{c}(0)^2 + 72q.} \tag{1}$$

For $q = 0, 0.1, 1$, the formal speeds are 1.90159, 3.28877 and 8.69575, while coarse generator/unitary/heat residuals remain below 2×10^{-15} .

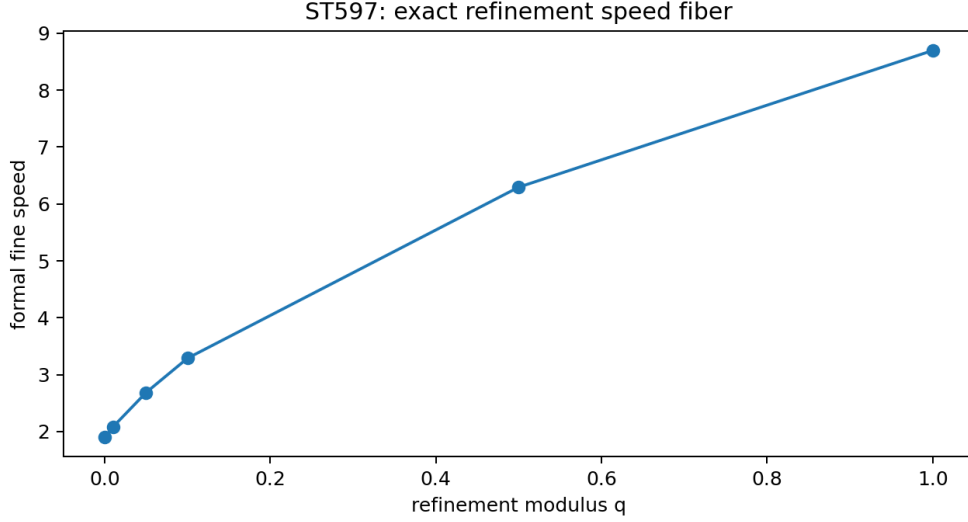


Figure 6: A continuous unbounded fine-speed fiber with identical complete coarse dynamics.

Theorem 6.1 (Coarse speed nonidentifiability). *No observable constructed solely from complete coarse spectral dynamics identifies q or the fine speed in (1).*

A single child-resolved odd Fourier mode recovers q because every odd eigenvalue shifts by exactly $2q$.

Trace-density conservation conditionally selects $q = 0$, but this is an added refinement axiom. Unlabelled-fiber naturality similarly leaves a vertical sector rate $v \geq 0$ invisible to coarse dynamics.

6.2 Round-IV ledger

Program	Status	Result
ST597	[PROVEN]	Exact speed fiber $\hat{c}^2 = C + 72q$.
ST598	[PROVEN]	Complete coarse observer is blind to fine speed.
ST599	[PROVEN]	Antipodal-rate speed-scaling trichotomy.
ST600	[CONDITIONAL]	Trace stationarity selects $q = 0$.
ST601–602	[PROVEN]	Hidden vertical sector and fine tomography of q .
ST603–605	[PROVEN]	Calibration compensation, locality cost and canonical-speed no-go.
ST606–608	open/gated	Transition, algebra and IR predictor chart obligations.
ST609	[PROVEN]	Dual dynamics does not derive propagation speed.
ST610–611	blocked	No physical speed/source/evidence closure.

7 Round V: ST612–ST626

7.1 Multilevel speed recurrence

For a dyadic tower with parent size N and new antipodal rate q_N ,

$$C_{2N} = C_N + \frac{N^2}{2} q_N. \quad (2)$$

Hence

$$C_j = C_0 + \frac{1}{2} \sum_{i < j} N_i^2 q_i.$$

For nonnegative rates, a finite infinite-depth speed exists exactly when

$$\sum_i N_i^2 q_i < \infty.$$

The often suggested law $q_N \sim \kappa N^{-2}$ adds $\kappa/2$ per layer and therefore diverges linearly with depth. Exact speed stationarity forces $q_N = 0$ at every layer.

Uniform exponential locality is stricter: a fixed exponential moment requires $q_N \sinh(\mu N)$ bounded. Every polynomial N^{-p} fails; exponential distance decay is necessary within this bound architecture.

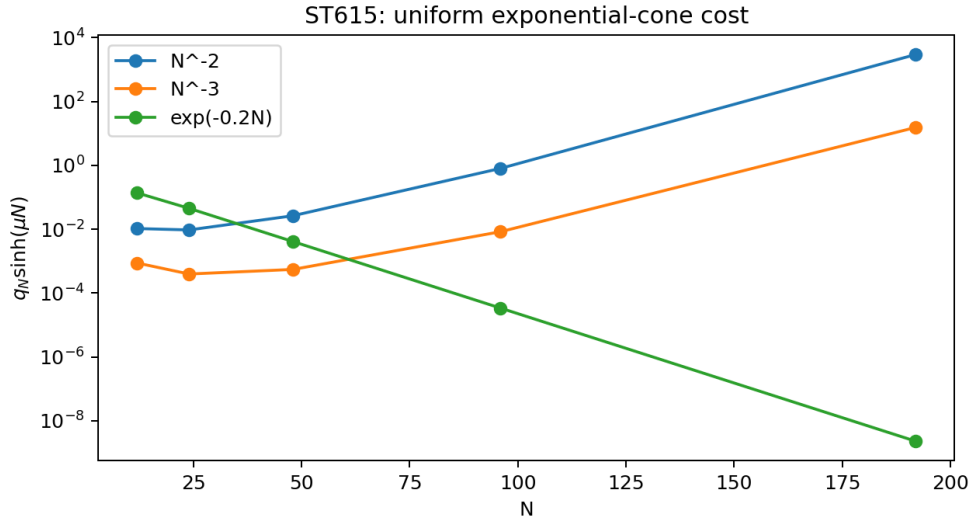


Figure 7: Polynomial antipodal rates fail a uniform exponential locality cost.

Two depth-three towers with final formal speeds 1.90159 and 4.47572 have identical base generator and heat records below 1.2×10^{-15} . Coassociativity does not constrain the one new rate per level. Each rate is nevertheless exactly recoverable by a child-resolved odd mode.

7.2 Round-V ledger

Program	Status	Result
ST612–614	[PROVEN]	Speed recurrence, convergence criterion and conditional zero tower.
ST615	[PROVEN]	Polynomial antipodal rates fail a uniform exponential cone.
ST616–617	[PROVEN]	Multilevel coarse blindness and level tomography.
ST618–619	[PROVEN]	Rate-record complexity and coassociativity no-selection.
ST620–621	[PROVEN]	Trace/speed laws share zero solution but remain inequivalent; calibration compensation.
ST622	[CONDITIONAL]	Zero-rate quadratic-symbol continuum candidate.
ST623	[PROVEN]	Refinement universality classes separated.

Program	Status	Result
ST624–626	blocked	No rate source, physical continuum or evidence.

8 Round VI: ST627–ST641

8.1 One-clock no-go

Let $a_N = L/N$ and keep physical wave number k fixed. The zero-rate finite-range symbol satisfies

$$\lambda_N(a_N k) = C a_N^2 k^2 + O(a_N^4 k^4).$$

Thus:

$$t_U, t_H \sim a_N^{-2}, \quad t_W \sim a_N^{-1}.$$

Theorem 8.1 (Single-clock no-go). *No single power-law clock $t \sim a^{-z}$ produces simultaneous nontrivial unitary/heat and second-order wave limits. The first pair requires $z = 2$, the wave channel requires $z = 1$.*

The minimal architecture contains one diffusive/Schrödinger clock class and one ballistic wave-clock class. Their absolute normalizations remain free.

8.2 Conditional continuum bridge

For the added zero-rate finite-range tower,

$$\frac{A_N}{a_N^2} \longrightarrow -C \partial_x^2, \quad C = 3.8153141155, \quad \hat{c} = \sqrt{C} = 1.9532829072.$$

The consistency error is $O(a_N^2)$ and controlled by the finite fourth moment 52.4176613.

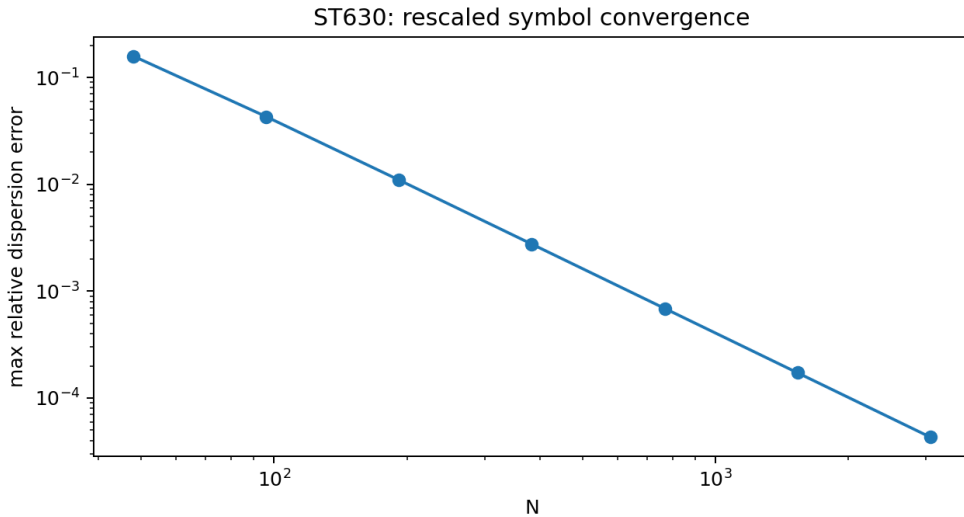


Figure 8: Convergence of the first three rescaled physical modes to Ck^2 .

For a localized displacement at physical time $T = 0.5$, the relative finite-lattice tail beyond $1.05\hat{c}T$ decreases to 8.20×10^{-9} at $N = 12288$.

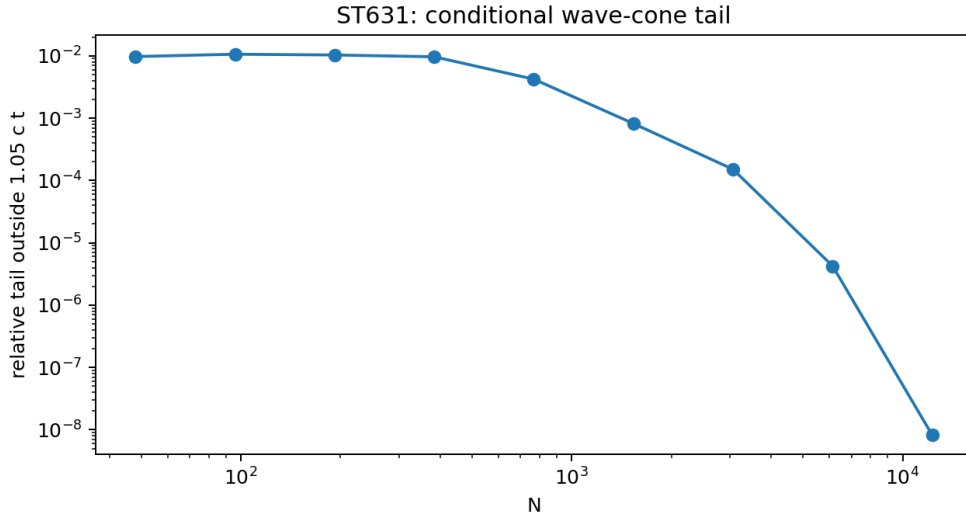


Figure 9: Conditional zero-rate wave tails vanish toward a continuum cone.

Under consistency and stability, the limiting scalar equation is

$$\partial_T^2 \phi - C \partial_x^2 \phi = 0,$$

with exact $1 + 1$ domain of dependence speed $\sqrt{C}$. This is the shortest current mathematical path from FIN spatial weights to a causal wave equation.

It is conditional. The zero-rate tower, coordinate embedding, ballistic clock and calibration are not generated by A_{12} . Moreover a scalar $1 + 1$ equation does not supply $3 + 1$ isotropy, Lorentz boosts, Maxwell fields, photons or matter coupling.

8.3 Round-VI ledger

Program	Status	Result
ST627–629	[PROVEN]	Scaling exponents, one-clock no-go and two clock classes.
ST630	[PROVEN]	Rescaled dispersion converges to Ck^2 .
ST631	[STRONG EVIDENCE]	Finite-lattice wave tails converge numerically toward zero.
ST632–633	[CONDITIONAL]	Continuum consistency and three functional calculi.
ST634	[PROVEN]	Duality survives; wave scaling separates.
ST635	[PROVEN]	Dimensionless speed plus one wave length/time anchor.
ST636	[CONDITIONAL]	Scalar continuum light-cone mechanism.
ST637	[PROVEN]	Light-cone slope is not Lorentz/light-field closure.
ST638	consolidated	No inevitability, but one coherent conditional bridge.
ST639–641	blocked	Independent frontiers, source package and evidence absent.

9 Falsification and obstruction ledger

Table 8: What failed and what follows.

Attempt	Result	Licensed conclusion
Read c directly from C_{12}	Formal, phase and group speeds disagree	No continuum speed at the end-point.
Positive $d^{-1.8}$ refinement	Fractional exponent 0.8	No finite nonzero low- k wave speed.
Literal signed strict extension	Negative modes at $N = 48$	Not a universal positive Laplacian refinement.
Coarse exact naturality	Continuous unbounded speed fiber	Complete coarse dynamics cannot identify fine speed.
$q_N \sim N^{-2}$ tower	Speed coefficient grows with depth	Per-layer finite contribution is not an infinite limit.
Polynomial antipodal locality	Exponential moment diverges	No uniform exponential cone from polynomial q_N .
One refinement clock	Requires both $z = 2$ and $z = 1$	At least two channel conversion classes.
Naive log interval chart	Negative Krawczyk margin	Method stop; numerical root not refuted.
Zero-rate local tower	Conditional PDE cone succeeds	Compatibility demonstrated, inevitability not.

10 Deepest interpretation after six rounds

Deepest surviving interpretation. FIN’s finite spectral core does not contain a unique speed of light. It contains enough structure to support a conditional local wave continuum once a refinement universality class, wave-clock scaling and calibration are selected. Exact coarse self-similarity hides rather than determines the fine propagation law.

The internal-observer intuition is mathematically viable in the following form:

$$c_n^{\text{phys}} = \frac{\ell_n}{\tau_n} \hat{c}_n = c_*.$$

But this equality is a constraint on three structures: the fine generator, the layer coordinate map and the calibration cocycle. A coarse observer may see identical dynamics while all three vary in the hidden fiber. Therefore constant internally measured c is not sufficient to reconstruct the external/fractal layer law.

The result is simultaneously positive and negative:

- **Positive:** a precise conditional sequence reaches a scalar continuum cone.
- **Negative:** current FIN does not select that sequence, one universal clock is impossible, and the observed SI value remains an anchor rather than a prediction.

11 Recommended next cycle: ST642–ST656

Table 9: Ranked next studies.

Rank	Programme	Acceptance criterion
1	ST642 zero-rate source theorem	Derive $q_N = 0$ from a new strict locality/action principle, or prove no invariant principle in a declared class can do so.
2	ST643 Green–Schur locality selector	Test whether response-preserving compression plus uniform locality forces the zero tower.
3	ST644 continuum convergence theorem	Add stability and prove strong convergence of wave propagators, not only stencil consistency.
4	ST645 uniform tail certificate	Replace numerical tails by an analytic bound uniform in N outside $(1 + \epsilon)ct$.
5	ST646 $3 + 1$ isotropic refinement	Construct or obstruct an isotropic carrier family with quadratic symbol.
6	ST647 emergent Lorentz boosts	Test convergence of discrete energy-momentum transformations and quantify boost-breaking corrections.
7	ST648 gauge/light sector	Determine the additional bundle/connection data needed for Maxwell rather than a scalar wave.
8	ST649 two-clock operational protocol	Freeze preparations and records that independently identify ballistic and diffusive clocks.
9	ST650 independent ruler/clock anchor	Design a noncircular calibration test; defining the ruler through light is rejected.
10	ST651 long-range polynomial cone	Prove the sharp causal-tail exponent for the positive $d^{-1.8}$ class.
11	ST652 signed oscillatory Krein route	Classify whether an indefinite refinement has a stable causal interpretation without pretending it is Markov.
12	ST653 predictor-correlated IR intervals	Retain z – b dependence symbolically to avoid the failed naive log interval.
13	ST654 transition/algebra state map	Reopen only with a genuinely new interval or characteristic-zero method.
14	ST655 strict source gate	Accept only a new formula-level refinement/clock/selector provider.
15	ST656 evidence gate	Accept only independently custodied laboratory records.

12 Final conclusions

1. The fixed strict operator has no exact causal cone and does not by itself possess a continuum wave number.
2. A finite nonzero low- k speed exists exactly for a quadratic generator symbol.
3. Positive $d^{-1.8}$ refinement is fractional; naive signed strict refinement loses positivity.
4. Exact dyadic refinement produces an unbounded fine-speed fiber invisible to every coarse spectral record.
5. Across a tower, finite speed requires $\sum N^2 q_N < \infty$; uniform exponential locality is stricter.
6. Exact speed stationarity conditionally selects the zero-rate tower, but the stationarity law is not strict-sourced.

7. One clock cannot preserve unitary/heat and wave limits. Two channel-specific clock classes are minimally required.
8. The zero-rate finite-range tower conditionally converges to a scalar $1 + 1$ wave operator with a causal cone.
9. This does not derive the SI value of c , Lorentzian $3 + 1$ spacetime, Maxwell light, apparatus or experiment.
10. Current FIN therefore supports a mathematically coherent route to causal waves but not a unique or inevitable physical bridge.

A Reproducibility inventory

The authoritative round outputs are

- `FIN_ST552_ST566_Results.json`,
- `FIN_ST567_ST581_Results.json`,
- `FIN_ST582_ST596_Results.json`,
- `FIN_ST597_ST611_Results.json`,
- `FIN_ST612_ST626_Results.json`,
- `FIN_ST627_ST641_Results.json`,

with six Python sources, six test modules and figure directories. The combined suite passes 31 tests.

B Nonclaims

This release does not complete legacy-to-strict, transfer legacy physical roles, discharge *QW*-2191, source gain/selector/units, derive a physical photon, Standard Model, gravity, L_{total} or ToE, or claim laboratory confirmation. All future research PDFs remain English.